Automatic Determination of the Indispensability Index of a Term Within a Specialized Domain

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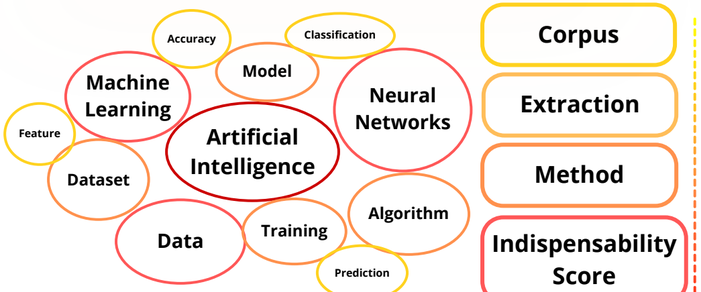
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AIRDIL Laboratoire Inter-universitaire de Recherche en Didactique Lansad (LANgues pour Spécialistes d'Autres Disciplines)

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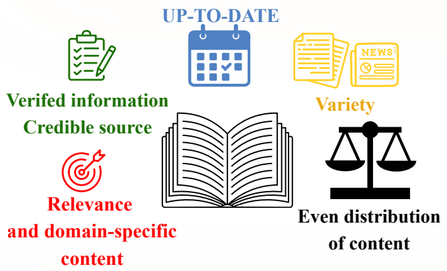
Introduction

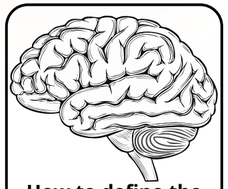


Corpus → Extraction → Method → Indispensability Score → Key Words

How can we determine a score that would reflect the importance of a term within a corpus?

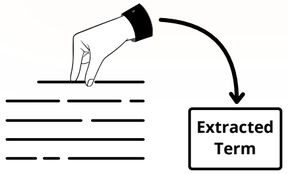
Quality Corpus: A Crucial Step





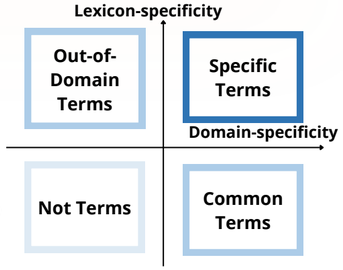
How to define the notion of indispensability?

Term extraction



- **Noise reduction**
- **Improving the relevance of the results**

- **Corpus dependent**
- **Difficulty** in identifying **rare** and **specific terms**



(Shang et al., 2017), (Xu et al., 2025)

Priorization methods

Statistical methods



Frequency in the document — **Rarity in the corpus**

Essential term

TF-IDF(t,d,D) = TF(t,d) x IDF(t,D)

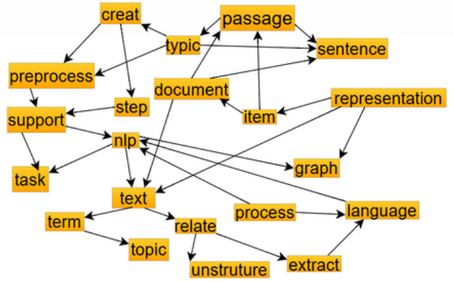
Frequency · Occurrence · Local weight | **Rarity · Global weight · Discrimination**

Advanced methods (TF-SYN-NER-Rel, YAKE, RAKE) incorporate **additional signals**.

- **Simple methods**
- **Limited semantic understanding**

(Salton et al., 1975), (Rose et al., 2009), (Zhou & Slater, 2003), (Rodionov et al., 2025)

Graph based approaches



Graph Construction → **Assign edge weight** → **Propagation of scores**

$$Score(w) = \Sigma\ [Score(v) / degre(v)] \text{ for each neighbor } v$$

- **Reveal structural roles**
- **Depend on graph construction**

(Malliaros & Skianis, 2015), (Ushio et al., 2021), (Zhang et al., 2024)

Hybrid methods

NBOW2:

Input text → Word embeddings → Computing of weights (TF-IDF) → Supervised learning → Key Words

SIFRank:

embeddings statistics



- **Take advantage of each type of method**
- **Depend on pre-trained embeddings**
- **Hight computational cost**

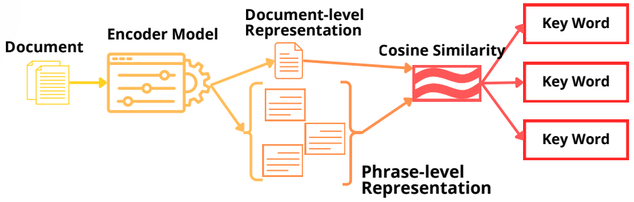
(Sheikh et al., 2016), (Giarelis & Karacapilidis, 2024)

Embeddings methods

- **Dense** embeddings: Word2Vec, GloVe and KeyBERT
- **Sparse** embeddings: SPLADE

[0.21, 0.74, 0.08, ..., 0.91, 0.44] — 768 dim (all non-zero)

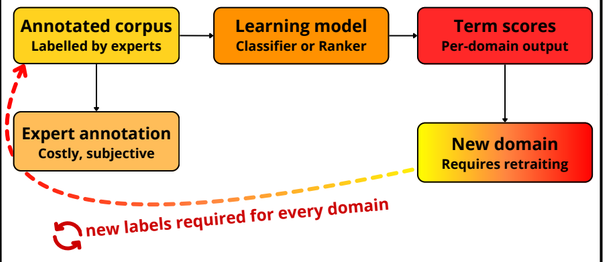
[0, 0, 0, 1, 0, 0, 0, 0.8, ..., 0, 0.3] — 30 522 dim (mostly zeros)



- **Placing semantically similar concepts close together.**
- **Difficulty of interpretation**

(Pennington et al., 2014), (Devlin et al., 2019), (Formal et al., 2021), (Giarelis & Karacapilidis, 2024)

Supervised methods

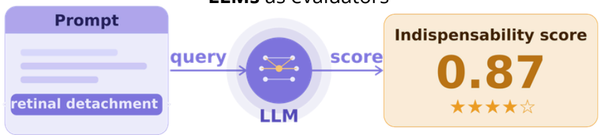


- **The model give a score from the learning of the labelled**
- **Assume a stable, human-annotated ground truth**

(Patra & Singh, 2013), (Sheikh et al., 2016)

Future work

LLMs as evaluators



Unified hybrid pipeline (statistics, graph, embeddings, supervised -> single interpretable scoring function)

